SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AIbasics such as intelligent agents and uninformed search methods to solve problemseffectively. (BL3)		
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Section / Batch:	2026/3rd	**Date:**	09.10.2026

Code of Conduct

I S.Yamini ,1973260015 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will resultin disciplinaryaction.*

Signature of the Student: S.Yamini

Instructions

- Readeachquestion carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Eachquestiongives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement,component, iteration or step number), explain its root cause, correct it with justification andapplythecorrected concept. An error merely carried forward counts once.
- Followtheconventions stated in each question (alphabetical expansion, explored set, goal test on removalfromthe frontier, tie-breaking) and present tables neatly.
- Marksfollowthe Error Analysis rubric: error identification, root cause analysis, correction and justification,application of concepts, and clarity and presentation.
- UploadinGitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation pageattheend).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches toAI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	
(ii)	
(iii)	

Answers – Question 1: Foundations of AI, Intelligent Agents and Rationality

(a) Correctness and erroneous claims

Statement	Status	Erroneousphrase / claim
S1	Incorrect	"internal reasoning always follows the laws of logic" and "thinking rationally". The Turing Test evaluates human-like conversational behaviour, not logical reasoning.
S2	Incorrect	AI winters did not occur in the 1950s before Dartmouth. The 1956 Dartmouth workshop introduced/coined the term AI; AI winters came later.
S3	Incorrect	"always achieves the best possible outcome" and "knows the actual result … in advance"; the taxi is wrongly called irrational after an unforeseeable landslide.
S4	Incorrect	PEAS categories are mixed up: steering/accelerator/brake/horn are actuators, while safe/fast/legal/comfortable are performance measures.
S5	Incorrect	Chess without a clock is fully observable (board visible), deterministic, sequential, static and discrete—not partially observable, stochastic or episodic.
S6	Correct	No erroneous claim. This accurately describes a model-based reflex agent.

(b) Root cause analysis

Error	Rootcause
S1	Confuses the Turing Test (acting humanly) with the laws-of-thought approach (thinking rationally).
S2	Confuses the beginning of AI as a named field with later periods of reduced funding and interest called AI winters.
S3	Confuses rationality with omniscience. Rational decisions depend on available percepts, prior knowledge, actions and expected outcomes, not perfect prediction.
S4	Does not distinguish an agent's goals/performance criteria, the external world, and the mechanisms that act on or sense the world.
S5	Misclassifies chess: the board and moves are observable and rule-governed; moves affect later moves, making it sequential. Without a clock, the environment is static.

(c) Corrected statements and justification

Statement	Corrected statement	Justification
S1	The Turing Test (1950) judges whether a machine's behaviour in conversation is indistinguishable from a human's. It is an example of the "acting humanly" approach.	The test evaluates observable behaviour, not whether internal reasoning follows formal logic.
S2	The 1956 Dartmouth workshop helped establish AI as a field and introduced the term "artificial intelligence". AI winters occurred later, when funding and interest declined.	The chronology in the original note is wrong.
S3	A rational agent selects the action expected to maximise its performance measure given its percept sequence and knowledge. A taxi choosing a route reported clear can still be rational if an unforeseeable landslide later blocks it.	Rationality is judged using information available at decision time, not hindsight.
S4	Performance measure: safe, fast, legal, comfortable trip. Environment: roads, traffic, pedestrians, weather. Actuators: steering, accelerator, brake, horn. Sensors: cameras, GPS, speedometer.	PEAS means Performance measure, Environment, Actuators, Sensors.
S5	Chess without a clock is fully observable, deterministic, sequential, static and discrete.	The board state is visible, rules determine outcomes, moves affect later states, no clock changes the state, and states/actions are discrete.
S6	No change.	The internal state uses a model of how the world evolves and how actions affect it, supporting action when a percept is incomplete.

(d) Application

(i) The taxi's choice was rational if it selected the route using the available traffic reports, its percept sequence, prior knowledge and the performance measure. The later unforeseeable landslide does not by itself make the earlier decision irrational.

(ii) The taxi environment is partially observable and dynamic, and it is uncertain/stochastic because events such as landslides cannot be predicted perfectly. It is also sequential and continuous in important aspects.

(iii) A simple reflex taxi might keep turning based only on the current camera view and repeatedly choose a blocked road because it has no memory of prior observations. A model-based reflex taxi keeps an internal state/model of the world, updated from percepts and knowledge of how the world and its actions change it.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

Astudent formulates the **MissionariesandCannibals**problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bankof a river, with a boat that carries one or two people and cannot cross empty. The boat is on the leftbank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank withno missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		
Note N1		
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error(component / note)	Root cause

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

Answers – Question 2: Problem-Solving Agents and Problem Formulation

(a) Correctnessof componentsandnotes

Component / Note	Status	What iswrong
State representation	Incorrect / incomplete	(M,C) omits the boat's bank. Use (M,C,B), where M and C are counts on the left bank and B is L or R.
Initial state	Incorrect / incomplete	Must include boat position: (3,3,L).
Actions	Incorrect / incomplete	The listed loads omit the boat's direction and legality conditions; actions must move one or two people from the bank where the boat currently is, and cannot move empty.
Goal test	Incorrect	M=C is not the goal. Goal is all people on the right bank and boat location is irrelevant: M=0 and C=0 on the left.
Legality check	Incorrect	Checks only the left bank. Must ensure on each bank missionaries are not outnumbered whenever missionaries are present.
Path cost	Incorrect	Objective is fewest crossings, so each crossing costs 1, not number of people carried.
N1	Incorrect	Correct order is goal formulation → problem formulation → search → execution.
N2	Incorrect	There are 4×4×2 = 32 syntactic states before legality constraints, not 16; the stated count of 10 legal states is also wrong.

(b) Root cause analysis

Error	Rootcause
State / initial state	A state must include every variable needed to determine possible actions and future outcomes, including the boat's side.
Actions	Actions must be applicable from the boat's current bank and obey boat capacity and crossing rules.
Goal test	The goal test must represent the requested final situation, not an equality between unrelated counts.
Legality	Safety constraints apply to both banks; checking only one side can admit unsafe states.
Path cost	The cost function must match the objective: minimise crossings, so every crossing has unit cost.
N1	A problem-solving agent defines goals and a problem before searching for a solution and executing it.
N2	State-space size must account for both counts (0–3 each) and the boat side, then filter by legality.

(c) Corrected formulation

Component	Correctedformulation	Justification
State	(M,C,B): number of missionaries and cannibals on the left bank; B∈{L,R}.	Captures the full situation.
Initial state	(3,3,L).	All six people and boat start on left.
Actions	Move 1M, 2M, 1C, 2C, or 1M+1C across from the boat's current bank, if available and legal.	Enforces capacity, direction and non-empty boat.
Goal	M=0 and C=0 (everyone on right bank).	All people have crossed.
Legality	On each bank, either M=0 or M≥C. Counts must remain between 0 and 3;	Checks safety on both banks.
Path cost	boat moves one or two people.	Minimises number of crossings.
N1	1 per crossing.	This is the problem-solving sequence.
N2	Goal formulation → problem formulation → search → execution. There are 32 possible encoded states (4 values of M × 4 values of C × 2 boat sides) before legality filtering.	The student omitted the boat-side dimension.

(d) Apply the corrected formulation

Boat load from (3,3,L)	Resultingstate(M,C,B)	Legal? Reason
1M	(2,3,R)	No. Left bank has 2M,3C; cannibals outnumber missionaries.
2M	(1,3,R)	No. Left bank has 1M,3C; unsafe.
1C	(3,2,R)	Yes. Left bank 3M,2C; right bank 0M,1C, safe because no missionaries there.
2C	(3,1,R)	Yes. Left bank 3M,1C; right bank 0M,2C, safe.
1M + 1C	(2,2,R)	Yes. Both banks have equal numbers of missionaries and cannibals.

(ii) Example: state (M=1, C=2, boat=R). The student's left-bank-only check may accept it because it checks only the left bank (depending on the given left-bank count); the right bank must also be checked. More directly, any state with M=1,C=2 on the left is rejected by the given left-bank rule, so the student's check's main defect is failure to check the right bank—for example (M=3,C=3, boat=R) is accepted by the left-bank check but the right bank has 0M,0C and is safe; a clearer illegal state is (M=2,C=2, boat=R) only if the right-bank counts violate safety. The legality

check should be applied to both banks in all cases.

(iii) An optimal solution has 11 crossings; with unit cost per crossing, the optimal path cost is 11.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

Astudent ran BFS, DFS and uniform-cost search (UCS) on theweighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A] [G,	{S, C, D, B} {S,
5	E	A]	C, D, B, E} {S,
6	G (goal)	[A]	C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	StackafterPushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S –D– G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	NodeExpanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			

Justification of the corrections	

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

Answers – Question 3: Searching for Solutions and Uninformed Search

(a) Errors in traces and remarks

Where	What is wrong
BFS iteration 2	Expanding C adds A, but A is not yet explored or queued; queue [D,A] is correct at this point.
BFS iteration 3	Expanding D must not add C again (already explored) and must not add A again (already in queue). Correct queue after this iteration is [A,B,E,G].
BFS iteration 6 / explored set	The goal test is on removal from the frontier. G is the goal and is removed; the explored set shown omits G only because the trace stops at goal detection.
BFS result	BFS finds the fewest-edge path S–D–G (2 edges), cost 9. It is a valid BFS path, but not necessarily minimum weighted cost.
DFS iteration 1	Alphabetically first child must be popped first. With stack top on left and the specified push convention, the stack after expanding S should be [C,D], not [D,C].
DFS trace/result	The corrected DFS follows S→C→A→B→D→E→G (with explored/stack duplicate rules), rather than immediately popping D and G.
UCS iteration 3 onward	From D (g=3), E has cost 5, B cost 7, and G cost 9; A already has cost 9 via C. The queue must retain/replace frontier entries consistently and process lowest g first.
R1	BFS graph search needs an explored set to prevent revisiting expanded nodes and cycles.
R2	Same path does not imply same optimality. BFS/DFS ignore weights; UCS minimises nonnegative path cost.
R3	Fewest frontier removals alone does not make DFS the best strategy; completeness, optimality, graph structure and cost objective matter.

(b) Root cause analysis

Error group	Root cause
BFS duplicate entries	Graph search must avoid adding nodes already explored or already in the queue; otherwise duplicates and incorrect queues arise.
Goal/explored handling	The specified goal test occurs when a node leaves the frontier; the trace must stop at that point.
BFS optimality	BFS minimises number of edges, not weighted cost.
DFS order	Stack order and child insertion must follow the stated alphabetical-first-pop convention.
DFS conclusion	DFS's first found goal depends on traversal order and is not guaranteed to be cheapest or shortest.
UCS trace	UCS orders the frontier by cumulative path cost g(n) and replaces a frontier entry when a cheaper path is found.
R1–R3	Explored sets avoid repeated work; optimality depends on the algorithm's cost assumptions, not just how quickly a trace terminates.

(c) Corrected traces

(i) BFS

Queue after iteration 3: [A, B, E, G].

Order dequeued (goal test on removal): S, C, D, A, B, E, G. The BFS path is S–D–G, with 2 edges and weighted cost 9.

Iter	Node dequeued	Queue after expansion	Explored set
. 1	S	[C,D]	{S}
2	C	[D,A]	{S,C}
3	D	[A,B,E,G]	{S,C,D}
4	A	[B,E,G]	{S,C,D,A}
5	B	[E,G]	{S,C,D,A,B}
6	E	[G]	{S,C,D,A,B,E}
7	G (goal)	[]	Goal test succeeds on removal

(ii) DFS

Using stack top on the left and alphabetical-first child popped first:

Iter	Node popped	Stack after pushing children
. 1	S	[C,D]
2	C	[A,D]
3	A	[B,D]
4	B	[D]

Iter	Node popped	Stack after pushing children
. 5	D	[E,G]
6	E	[G]
7	G (goal)	[]

(iii) UCS
Priority queue entries are shown as Node:g(parent); ties are alphabetical.

Iter	Node expanded	g(n	Priority queue after expansion
. 1	S	) 0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	–

Justification: BFS avoids duplicate frontier/explored nodes and explores by depth. DFS follows the specified stack order and may find a nonoptimal path. UCS always removes the least-cost frontier node; the route S–D–E–G costs 3+2+3 = 8 and is cheaper than S–D–G (9).

(d) Comparison and explored-set application

Strategy	Path found	Path cost	Nodes removed	Optimal here?
BFS	S–D–G	9	7	Shortest by number of edges; not minimum weighted cost.
DFS	S–C–A–B–D–E–G	19	7	No
UCS	S–D–E–G	8	6	Yes

(ii) At BFS iteration 3, D's neighbours include C, which is already explored, and A, which is already in the queue from expanding C. The explored set and frontier check prevent duplicate visits and keep the search graph orderly.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problemformulation,searchwithpartialinformation(sensorlessandpercept-basedbelief-state search)

Avacuum agent lives in a two-squareworld(Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where(task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	BeliefState after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	
2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	AfterPercept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

Answers – Question 4: Search with Partial Information

(a) Errors in tables and conclusions

Where	What iswrong
Task A, step 1	After Suck from {S1,S3,S5,S7}, the correct belief state is {S5,S7}; this row is correct.
Task A, step 2	Moving Right from states with agent at L produces {S6,S8}; correct.
Task A, step 3	Suck at R cleans the right square in S6 but S8 is already clean, so the resulting belief state is {S8}; the student incorrectly leaves {S6,S8}.
C1	A belief state can contain multiple physical states yet a sensorless plan can still guarantee the goal if every possible state in the belief set reaches a goal state.
Task B, step 0 update	Percept [Left, Dirty] should retain only states with agent at L and left square dirty: {S1,S3}; student incorrectly includes S2 and
Task B, step 1 predicted	S4.
Task B, step 2 predicted/update	After Suck from {S1,S3}, predicted belief is {S5,S7}; not {S4,S5,S7,S8}.
C2	After Right from {S5,S7}, predicted is {S6,S8}; percept [Right, Dirty] filters this to {S6}, which is correct if the prior belief is correct.
C3	{S6} means agent is at Right and the right square is Dirty; goal has not been reached because both squares are not clean.

(b) Root cause analysis

With 8 physical states, there are 2^8 = 256 possible belief states (all subsets), not 8.

Error group	Root cause
Task A step 3	The transition model for Suck was not applied to the current square: at R, Suck cleans the right square.
C1	Confuses knowing the exact physical state with guaranteeing a goal. Sensorless planning reasons over all states in a belief set.
Task B step 0	A percept updates belief by filtering out every state inconsistent with the reported location and current-square status.
Task B step 1	Predicted belief must be computed by applying the action transition to each state remaining after the prior percept.
C2	Confuses knowing location/status with satisfying the goal condition that both squares are clean.
C3	A belief state is any subset of physical states, so the number of subsets grows exponentially.

(c) Corrected tables and conclusions

(i) Task A – sensorless belief states

Step	Action	Belief state after action
0	Start	{S1,S3,S5,S7}
1	Suck	{S5,S7}
2	Right	{S6,S8}
3	Suck	{S8}

(ii) Task B – action prediction and percept update

Step	Action	After action (predicted)	Percept	After percept (updated)
0	Start	{S1,S2,S3,S4,S5,S6,S7,S8}	[Left, Dirty]	{S1,S3}
1	Suck	{S5,S7}	[Left, Clean]	{S5,S7}
2	Right	{S6,S8}	[Right, Dirty]	{S6}

(iii) Corrected conclusions

C1: Incorrect. A sensorless problem can be solved when a plan takes every state in the initial belief set to a goal state. For Task A, [Suck, Right, Suck] reaches {S8}, where both squares are clean, so it guarantees the goal.

C2: Incorrect. {S6} represents (R,C,D): the agent is on the right, the left square is clean, and the right square is dirty. The goal is not reached.

(d) Apply the corrected concepts

(i) There are 2^8 = 256 possible belief states because each of the 8 physical states may either be included or excluded from a subset. Belief-state search is harder because its state space is exponential in the number of physical states.

(ii) Initial belief {S1,S3,S5,S7}. After **Right**: {S2,S4,S6,S8}. After **Suck**: {S2,S4,S8} (S2 and S4 remain dirty on the left; S8 is clean on both squares). This is not a solution because not every possible state in the belief set satisfies the goal.
Note: In Task B, the percept update at step 1 leaves {S5,S7}; after Right and the [Right, Dirty] percept, the updated belief is {S6}.

Rubrics forError Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____